Research Note: Amplitude Estimation and Tensor-Network Baselines in QF-AgentOS

QF-AgentOS — An honest benchmark and evidence harness for quantum finance

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Two research-grade quantum-finance extras were added to QF-AgentOS, each chosen because it sharpens the platform's core question — *does quantum actually help?* — rather than blurring it. This note records the methodology, the honest complexity analysis, and the verdict for both.

Part A — Quantum Amplitude Estimation for risk

Implementation: `qf_agentos/finance/qae.py`; CLI: `qf-agent estimate`.

1. Problem statement

Estimate an expectation under a discretised distribution:

$$\mathbb{E}[f(X)] = \sum_{i=0}^{2^m-1} p_i\, f_i, \qquad f_i \in [0,1].$$

Concrete instances shipped: the **expected loss** of a discretised truncated-normal loss distribution over 2^m levels, and a **VaR-style exceedance probability** $\Pr[\text{loss} > t]$ (payoff = tail indicator). Both are the canonical targets of quantum risk analysis (Woerner & Egger 2019) and derivative pricing (Stamatopoulos et al. 2020).

2. Classical baseline

Best classical method: For the small discretised distributions a statevector can hold ($m \leq\sim 12$), the expectation is an **exact sum** — $O(2^m)$, instant, zero error. For continuous/high-dimensional versions the workhorses are **Monte Carlo** ($O(1/\epsilon^2)$ samples for RMSE ϵ) and its stronger cousins **quasi-MC** (Sobol) and **multilevel MC** (Giles) — all production-scale. **Why this is the right baseline:** any quantum claim must beat *exact summation* at these sizes and *MC/QMC/MLMC* asymptotically — not a naive comparator (guardrail G-baseline).

3. Quantum approach

Primitive: Quantum Amplitude Estimation, specifically **Maximum-Likelihood AE** (Suzuki et al. 2020) — chosen over canonical (QPE-based) QAE because it avoids deep phase-estimation circuits, and over a raw single-shot estimate because MLAE resolves the amplitude-aliasing across Grover powers.

```
Algorithm: MLAE for E[f]
Input: probabilities p, payoff f in [0,1], Grover powers {k}, shots N
Output: estimate of a = E[f]
1. Build A: |0> -> sum_i sqrt(p_i)|i>(sqrt(1-f_i)|0> + sqrt(f_i)|1>)   # good-state prob == a
2. Grover operator Q = -A S_0 A^dagger O   (O marks objective qubit = |1>)
3. For each power k: P_k = <good| Q^k A |0> = sin^2((2k+1)theta);  h_k ~ Binomial(N, P_k)
4. theta_hat = argmax_theta sum_k [ h_k log sin^2((2k+1)theta) + (N-h_k) log cos^2(...) ]
5. return a_hat = sin^2(theta_hat)
```

The amplitude oracle is built and **validated exactly**: the objective-qubit good-state probability equals $\sum_i p_i f_i$ to machine precision (a unit test enforces this). The MLAE loop uses a real Grover operator on the statevector; only the shot sampling is stochastic.

4. Complexity analysis

Cost component	Classical (MC)	Quantum (QAE)
Per-query	$O(1)$ sample	$O(1)$ oracle call
State preparation	—	$O(2^m)$ (arbitrary distribution)
Queries for RMSE ϵ	$O(1/\epsilon^2)$	$O(1/\epsilon)$
Readout	—	$O(1)$ (single amplitude)
End-to-end	$O(1/\epsilon^2)$	$O(2^m + 1/\epsilon)$

Speedup status: **Proven** quadratic in queries (Brassard, Høyer, Mosca, Tapp 2002; Montanaro 2015). **Assumptions:** amplitude-oracle access — i.e. an efficient state-preparation unitary. For an *arbitrary* financial distribution this costs $O(2^m)$ and is the binding constraint (guardrail G2/G3): the quadratic advantage survives only when state prep is amortised across many pricings (same distribution, many strikes) (Stamatopoulos et al. 2020) or the distribution is efficiently loadable (Grover–Rudolph for log-concave; qGAN, Zoufal et al. 2019).

5. Hardware feasibility

Required regime: early fault tolerance. The quadratic advantage requires large Grover powers $\to$ deep *coherent* circuits; NISQ noise destroys them at useful precision. **Circuit depth:** $\propto \max_k(2k+1)\cdot\text{depth}(A)$, with depth$(A)$ itself $O(2^m)$ for exact loading. **Realistic horizon:** the threshold for a genuine advantage in derivative pricing is estimated at thousands of logical qubits and very deep circuits (Chakrabarti et al. 2021) — a 10+ year horizon.

6. Empirical result (this implementation)

On a 4-qubit expected-loss instance (`qf-agent estimate --qubits 4`): MLAE recovers the exact expectation to $\sim 10^{-3}$, but **classical MC at the same query budget matches or beats it**, and exact summation ($2^4 = 16$ terms) is instant and error-free. The resource analysis reports the proven $1/\epsilon$ vs $1/\epsilon^2$ ratio while flagging the $O(2^m)$ state-prep bottleneck. **Verdict: CLASSICAL PREFERRED** at every size the platform can run.

7. Open questions

1. State preparation competitive with the QAE inner loop (closing the end-to-end gap).
2. QAE + multilevel-MC coupling — can they compose below either alone?
3. Error-mitigated QAE that preserves the quadratic scaling.

Part B — Tensor-network classical-simulability baseline

Implementation: `qf_agentos/finance/tensor_network.py`*; CLI:* `qf-agent simulability`*.*

1. Problem statement

Given the QAOA output state $|\psi\rangle$ for a reduced portfolio/collateral/routing QUBO, decide: **is this quantum circuit classically simulable by a tensor network?** If a matrix-product state (MPS) of modest bond dimension χ reproduces $|\psi\rangle$, then a classical algorithm with $O(n\chi^2)$ resources matches the circuit — undercutting any quantum-advantage claim from that instance.

2. Classical baseline

Method: MPS / DMRG-style simulation. An n-qubit state has an exact MPS with bond dimension up to $2^{\lfloor n/2 \rfloor}$; the *useful* regime is when the entanglement across every cut is low, so χ stays small and simulation is polynomial (Vidal 2003). This is simultaneously a classical *competitor* and a diagnostic.

3. Method

For each bipartition, compute the Schmidt spectrum (SVD of the reshaped amplitude tensor), the bipartite von-Neumann entropy $S = -\sum_i \lambda_i^2 \log_2 \lambda_i^2$, and the bond dimension χ_ϵ needed to capture fidelity $1-\epsilon$. Then reconstruct a truncated bond-χ MPS and report its fidelity $|\langle \psi_{\text{MPS}}|\psi\rangle|^2$. Validated on product ($S = 0, \chi = 1$), Bell/GHZ ($S = 1, \chi = 2$), and random ($\chi = 2^{n/2}$ for exact recovery) states.

4. Complexity

Representation	Parameters	Simulable when
Statevector	2^n	always (small n)
MPS (bond χ)	$O(n\chi^2)$	$\chi \ll 2^{n/2}$ (low entanglement)

5. Empirical result

The shipped dense-QUBO instances produce **highly entangled** QAOA states — e.g. the 13-qubit collateral instance has max bipartite entropy ≈ 5.2 bits and needs $\chi \approx 43$ against an exact-rank max of 64. So an MPS gives *no* compression here; the analysis reports this honestly as a **small-instance artefact, not hardware advantage** (exact statevector already solves it classically). The value delivered is a *quantified* entanglement/compressibility measure and an honest crossover statement: MPS-simulable (low-χ) circuits are trivially classical; high-χ ones still fall to exact statevector at these sizes.

6. Caveats

- Dense (all-to-all) QUBOs generate near-maximal entanglement — MPS is the wrong classical tool for them; the honest baseline there is exact enumeration / MILP (already the platform's).
- The analysis uses the exact statevector, so it *quantifies* rather than *replaces* classical simulation — which is the intended, honest role.

Overall conclusion

Neither extra produces a quantum advantage on any instance QF-AgentOS can run — and both are engineered to *say so with evidence*. QAE's advantage is proven but asymptotic and state-prep-bottlenecked (early-FT horizon); the tensor-network baseline quantifies exactly when a QAOA circuit is classically simulable. Both are faithful to the platform's thesis: an agent that runs the quantum method honestly and reports, with proof, that classical still wins.

References

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Research artifact — decision-support only, not investment advice.